PORTFOLIO PROJECT · DATA PIPELINE · MARKET ANALYTICS

# Northeast Heating & Cooling Demand Map

Where and when the Northeast needs heating oil, propane and AC service, estimated county by county and month by month from public data

COMPLETE · 3 DASHBOARDS LIVE

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Python · SQL · Tableau

WHY THIS PROJECT

# Fuel dealers run a seasonal business with two peaks

### Winter

`heating oil · propane`

Fuel delivery volume spikes with cold weather. The Northeast is the heating oil capital of the U.S., and propane is common in rural areas.

### Summer

`AC install · repair`

Fuel slows down, but full-service dealers keep crews busy installing and repairing air conditioning.

### The gap

`no public dataset`

Nobody publishes monthly fuel demand by county. Software vendors serving these dealers have to guess where the demand is.

**Goal:** build that view from public data, so dispatch, routing and billing platforms can see where demand is, how it moves through the year, and which dealers fit best.

SCOPE

# Every county in the nine Northeast states, every month

**9**

states: CT, MA, RI, VT, NH, ME, NY, NJ and PA

**218**

counties, each with modeled monthly demand

**3**

demand types: heating oil, propane and an AC cooling index

**5**

recent heating seasons at monthly grain

**Plus a Connecticut pilot:** a dealer layer that maps fuel dealers against county demand and ranks them by fit for a full-service software platform.

PIPELINE

# Nine sources, one schema, one dashboard

### Sources

Census · EIA · NOAA

Heating fuel by home, usage, weather, prices and dealers.

### Extract

requests · BeautifulSoup

API pulls, file downloads and a dealer scrape, one script per source.

### Model

pandas · geopandas

Standardize FIPS codes, convert units, estimate demand and calibrate.

### Store + analyze

SQL

Star schema, nine analysis queries and Tableau-ready views.

### Visualize

Tableau Public

County maps with a month slider, rankings and KPIs.

**Sources:** Census ACS and TIGER/Line · EIA RECS, SEDS, SHOPP and 861M · NOAA degree days

DEMAND MODEL

# Homes × usage, shaped by the weather

```
est_gallons(county, fuel, month) =
  homes(county, fuel)
  × [ gal_per_hdd(state, fuel) × HDD(county, month)
    + base_gal(state, fuel) / 12 ]
```

### Homes

Occupied homes heating with oil or propane, from Census ACS.

### Usage

RECS heating gallons per home, divided by 2020 degree days, plus a steady base.

### Weather

Each county's own NOAA degree days, so cold winters burn more.

### Checks

Priced with EIA SHOPP, then calibrated against EIA SEDS state totals.

DELIVERABLES

# From a database to answers a sales team can use

## Star schema

SQLite · 5 dimensions · 8 facts

- About 140,000 rows, checked for broken links on load
- Demand, cooling, weather, prices and EIA totals

## 9 SQL analyses

CTEs · window functions · all 9 built

- Top counties, peak months, winters vs. normal
- Price volatility and who heats with what
- Dealer density and fit score (CT)

## Tableau dashboards

Tableau Public · live

- The Thermal Year: county map with a month slider
- Size vs. intensity: year-round opportunity
- Who to call first: Connecticut dealer prospects

**Year-round score:** 60% heating spend percentile + 40% cooling index percentile, to find the best markets for full-service dealers. Tested: the weights barely change the ranking.

EARLY RESULTS · 2025-26 SEASON

# A $16.8 billion market that peaks every January

**$16.8B**

estimated residential heating oil and propane spend

**3.2B**

gallons of heating oil, plus 1.0B gallons of propane

**Jan**

peak month in all 218 counties; Dec to Feb is half the year

**$686M**

largest county market: Suffolk County, NY

**Size vs. intensity:** ranked on totals, the best year-round markets are the big NYC and Boston counties. Per home, it is oil-heavy areas with warm summers: Sussex NJ, Northeastern CT, Putnam NY.

CONNECTICUT PILOT · 468 LICENSED DEALERS

# Who to call first

**468**

licensed heating fuel dealers, from the state license registry

**20%**

also hold an HVAC contractor license: full-service

**2,206**

oil and propane homes per dealer in Western CT, vs. 922 in the Lower River Valley

**#1**

Densmore Oil, Stonington: full-service, 22 years licensed, under-served region

**Fit score:** 50% offers HVAC service + 30% how under-served its region is + 20% years licensed. Built in SQL with window functions, ranked with ROW_NUMBER().

STATUS

# All seven phases are done

01 · DONE

### Planning

Questions, sources, method and schema documented

02 · DONE

### Data collection

Census, NOAA and EIA pulls in Python

03 · DONE

### Modeling

Demand model, calibration and sensitivity

04 · DONE

### SQL

Star schema, load and nine analyses

05 · DONE

### Tableau

Three dashboards live on Tableau Public

06 · DONE

### CT dealers

Dealer list, fit score and prospect ranking

07 · DONE

### Findings

Write-up and dashboard screenshots

**Honest limits:** county demand is modeled from state averages and weather, not measured. It shows patterns and relative size well, not exact gallons.